```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from google.colab import drive

drive.mount('/content/drive')

path = "/content/drive/MyDrive/weight of copper - weight of copper.csv"

df = pd.read_csv(path)
df.head()
```

```
Mounted at /content/drive
```

	Unnamed: 0	power_rating	weight_of_copper	category
0	0	16.46	34.69	high
1	1	12.52	28.78	high
2	2	7.66	21.48	high
3	3	21.09	41.64	high
4	4	24.43	46.64	high

```
df = df.drop("Unnamed: 0",axis = 1)
df.head()
```

	power_rating	weight_of_copper	category
0	16.46	34.69	high
1	12.52	28.78	high
2	7.66	21.48	high
3	21.09	41.64	high
4	24.43	46.64	high

```
K = {'low':0,'high':1}
df['category'] = df['category'].map(K)
X = df[["power_rating","weight_of_copper"]]
Y = df[["category"]]
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.33,random_state=42)
from sklearn.ensemble import RandomForestClassifier
clf = RandomForestClassifier(max_depth=2, random_state=0)
clf.fit(X_train, y_train.values.ravel())
```

```
▾ RandomForestClassifier
RandomForestClassifier(max_depth=2, random_state=0)
```

```
y_pred = clf.predict (X_test)
y_pred
```

```
array([1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1])
```

```
from sklearn.metrics import mean_squared_error
from sklearn.metrics import accuracy_score
print("accuracy score",accuracy_score(y_test, y_pred))
print("mean_squared_error",mean_squared_error(y_test, y_pred))
from sklearn.metrics import confusion_matrix
```

```
accuracy score 1.0
mean_squared_error 0.0
```

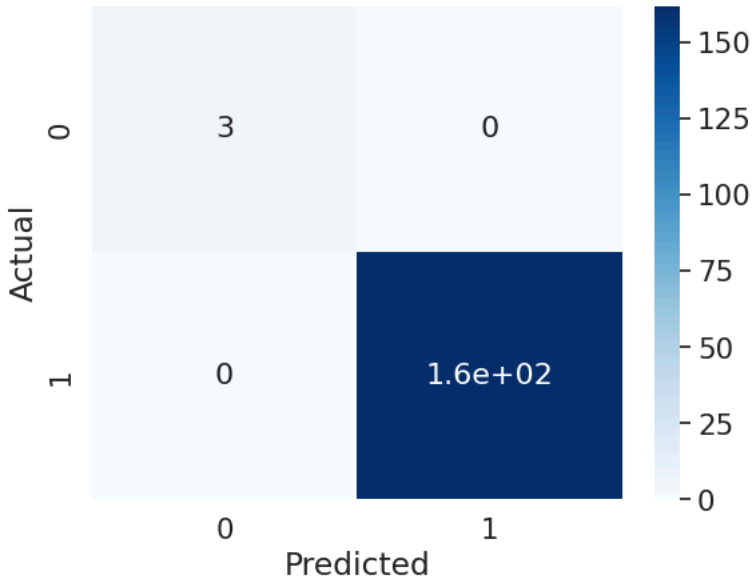
```
y_true = y_test
data = confusion_matrix(y_true, y_pred)
data
```

```
array([[  3,   0],
       [  0, 162]])
```

```
import seaborn as sn
df_cm = pd.DataFrame(data, columns=np.unique(y_true), index =np.unique(y_true))
df_cm.index.name = "Actual"
df_cm.columns.name = "Predicted"
plt.figure(figsize = (10,7))
sn.set(font_scale=1.4)
df_cm
```

Predicted	0	1
Actual		
0	3	0
1	0	162

```
<Figure size 1000x700 with 0 Axes>
```

```
sn.heatmap(df_cm, cmap="Blues", annot=True,annot_kws= {"size":16})
```

```
<Axes: xlabel='Predicted', ylabel='Actual'>
```